Machine Learning Model Project

Student Performance and Customer Purchase Behavior

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CSCI-164

Problem

- Predict Student Success (pass/fail)
- Predict customer purchase behavior
- Compare model performance across datasets
- Evaluate how model performance changes across datasets

Datasets

01 **Student Performance**

- Study time, absences, demographic
- More Structured
- Target: pass/fail

More Structured Dataset

02 **Customer Purchase**

- Customer behavior features
- More complex
- Target: purchase decision

More Complex Dataset

Preprocessing

- Encoding categorical variables
- Scaled numerical features
- Handled missing data
- Removed G1 and G2 (data leakage)
- Prevents overfitting
- Ensures fair model comparison

Models Used

- Logistic Regression → Linear model
- K-Nearest Neighbors (KNN) → distance-based
- Multi-Layer Perceptron (MLP) → non-linear

Training and Tuning

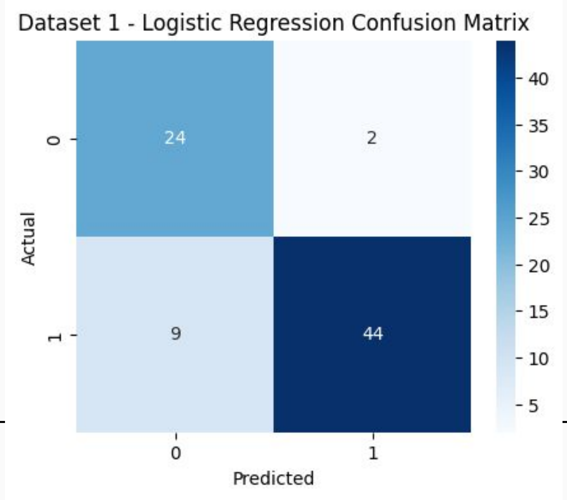
- 80/20 train-test split
- GridSearchCV for tuning
- Evaluated using accuracy, F1, ROC-AUC
- Cross-validation improves reliability
- Tuning optimizes model performance

Student Performance Results

- **Logistic Regression: 86.1% (best)**
- MLP: 82.3%
- KNN: 72.2%
- ROC-AUC highest: 0.94

- Data shows strong linear relationships
- KNN performs worse due to sensitivity to feature scaling and noise

	Model	Accuracy	Precision	Recall	F1 Score	ROC-AUC
0	Logistic Regression	0.860759	0.956522	0.830189	0.888889	0.942671
2	MLP	0.822785	0.897959	0.830189	0.862745	0.899129
1	KNN	0.721519	0.754098	0.867925	0.807018	0.780116

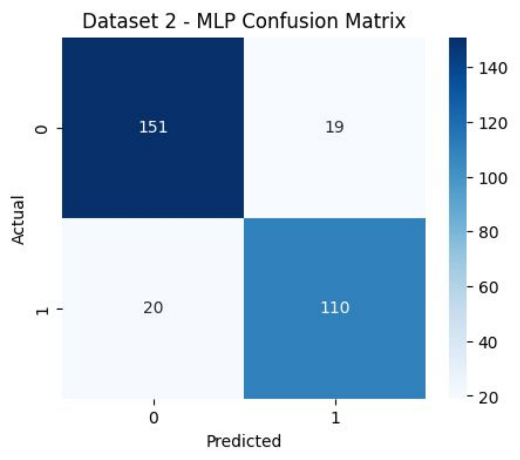


Customer Purchase Results

- **MLP: 87.0% (best)**
- KNN: 85.0%
- Logistic Regression: 82.7%
- ROC-AUC highest: 0.93

- MLP captures complex, non-linear patterns
- KNN performs well due to flexible decision boundaries

	Model	Accuracy	Precision	Recall	F1 Score	ROC-AUC
2	MLP	0.870000	0.852713	0.846154	0.849421	0.925023
1	KNN	0.850000	0.814815	0.846154	0.830189	0.902398
0	Logistic Regression	0.826667	0.825000	0.761538	0.792000	0.899548



Model Comparison

Key Comparisons

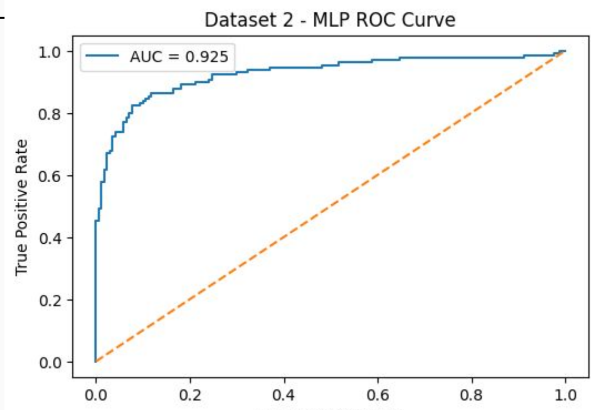
- Student Dataset → **Logistic Regression**
- Customer Dataset → **MLP**
- KNN performance varies across datasets

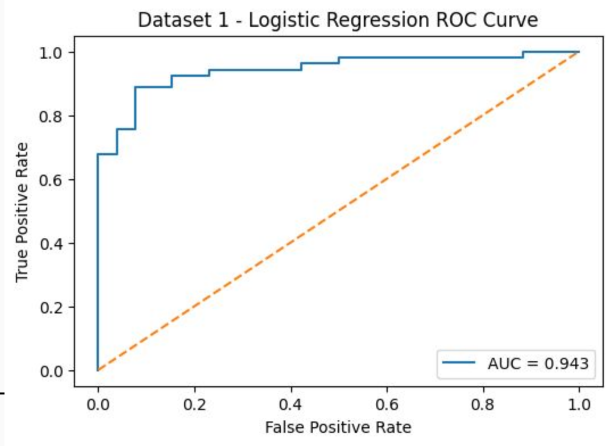
Key Insight

- Model performance depends on dataset complexity
- Simpler models work well on structured data
- Complex models perform better on non-linear data

	Model	Accuracy	Precision	Recall	F1 Score	ROC-AUC	Dataset
5	MLP	0.870000	0.852713	0.846154	0.849421	0.925023	Customer Purchase
4	KNN	0.850000	0.814815	0.846154	0.830189	0.902398	Customer Purchase
3	Logistic Regression	0.826667	0.825000	0.761538	0.792000	0.899548	Customer Purchase
0	Logistic Regression	0.860759	0.956522	0.830189	0.888889	0.942671	Student Performance
2	MLP	0.822785	0.897959	0.830189	0.862745	0.899129	Student Performance
1	KNN	0.721519	0.754098	0.867925	0.807018	0.780116	Student Performance

ROC Curve Insights





- Logistic Regression → best for Student dataset **(AUC ≈ 0.94)**
- MLP → best for Customer dataset **(AUC ≈ 0.93)**
- High AUC values indicate strong class separation
- Results confirm model comparison findings

Key Takeaways

- Model performance depends on dataset characteristics
- Logistic Regression performs well on structured, linear data
- MLP performs better on complex, non-linear data
- KNN performance varies depending on the dataset
- Hyperparameter tuning can improve performance but not guarantee

Conclusion

- Compared 3 models across 2 datasets
- Logistic Regression performed best for Student Dataset
- MLP performed best for Customer dataset
- Model performance depends on the dataset
- No single model is best for all problems